

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2024****Branch: Civil Engineering****Semester : IVth Semester****Course: B.Tech****Subject Code & Name: BTCVC406/BTCIC406/BTCEC406 Engineering Geology****Max Marks: 60****Date: 02/01/2025****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	What is the main focus of physical geology?	(BL1/CO1)	1
	a. Study of minerals b. Study of Earth's physical processes c. Study of fossils d. Study of underground water		
2	Which feature is created by wind erosion?	(BL1/CO1)	1
	a. Mushroom rock b. Moraine c. Meander d. Alluvial fan		
3	What causes metamorphism in rocks?	(BL1/CO2)	1
	a. Heat and pressure b. Cooling of magma c. Erosion and deposition d. Sedimentation		
4	What term refers to the way light reflects off a mineral's surface?	(BL1/CO2)	1
	a. Cleavage b. Hardness c. Lustre d. Specific gravity		
5	What type of fold is characterized by upward bending layers?	(BL1/CO3)	1
	a. Anticline b. Syncline c. Monocline d. Overlap fold		
6	Which zone lies above the water table?	(BL1/CO3)	1
	a. Unsaturated zone b. Saturated zone c. Aquifer d. Aquiclude		

7	What is the primary purpose of a preliminary geological survey?				(BL2/CO4)	1
	a. To assess geological conditions before construction	b. To classify rocks in a region	c. To explore mineral resources	d. To map rivers and lakes		
8	Which alternative method is sometimes used alongside drilling for better data?				(BL2/CO4)	1
	a. Geophysical surveys	b. Satellite imaging	c. Aerial photography	d. Remote sensing		
9	What is the primary tool used to collect cores in exploratory drilling?				(BL1/CO4)	1
	a. Core barrel	b. Auger	c. Rotary drill	d. Percussion drill		
10	Why is the Deccan Trap region significant for dam construction?				(BL2/CO5)	1
	a. It has basaltic rocks that are strong and impermeable.	b. It has a flat terrain ideal for reservoirs.	c. It is free from folded strata.	d. It has soft rocks suitable for excavation.		
11	What method is used to improve tunneling stability in fractured zones?				(BL2/CO5)	1
	a. Shotcreting and rock bolting	b. Core drilling	c. Soil compaction	d. Excavation blasting		
12	What is a geological consideration for dam alignment?				(BL2/CO5)	1
	a. Stability of rock strata	b. Presence of vegetation	c. Water flow speed	d. Soil color		
Q. 2	Solve the following.					12
A)	What are the primary and secondary effects of volcanic activities?				(BL2/CO1)	6
B)	Write a note on the earth's interior with a neat labeled diagram.				(BL2/CO1)	6

Q.3	Solve the following.		12
A)	Explain the difference between cataclastic and thermal metamorphism.	(BL4/CO2)	6
B)	Enumerate the procedure for the formation of sedimentary rocks.	(BL3/CO2)	6
Q.4	Solve Any Two of the following.		12
A)	Define fault and explain its civil engineering significance.	(BL4/CO3)	6
B)	Differentiate between types of unconformities based on their geological characteristics.	(BL4/CO3)	6
C)	Differentiate between confined and unconfined aquifers with examples.	(BL4/CO3)	6
Q.5	Solve Any Two of the following.		12
A)	Provide a detailed explanation of preservation core lithologs.	(BL3/CO4)	6
B)	Define geological investigations and list the primary steps involved in them.	(BL2/CO4)	6
C)	What is exploratory drilling? Describe its purpose in geological investigations.	(BL2/CO4)	6
Q.6	Solve Any Two of the following.		12
A)	Write a detailed note on various tunnel excavation techniques.	(BL3/CO5)	6
B)	Define dam and describe various types of dam structures.	(BL3/CO5)	6
C)	Explain various problems faced in the tunnelling construction in folded and faulted strata.	(BL4/CO5)	6
	*** End ***		